

JAL230I DATA SCIENCE

UNIT III

DESCRIPTIVE STATISTICS - DESCRIBING RELATIONSHIPS

UNIT III DESCRIPTIVE STATISTICS - DESCRIBING RELATIONSHIPS - SYLLABUS

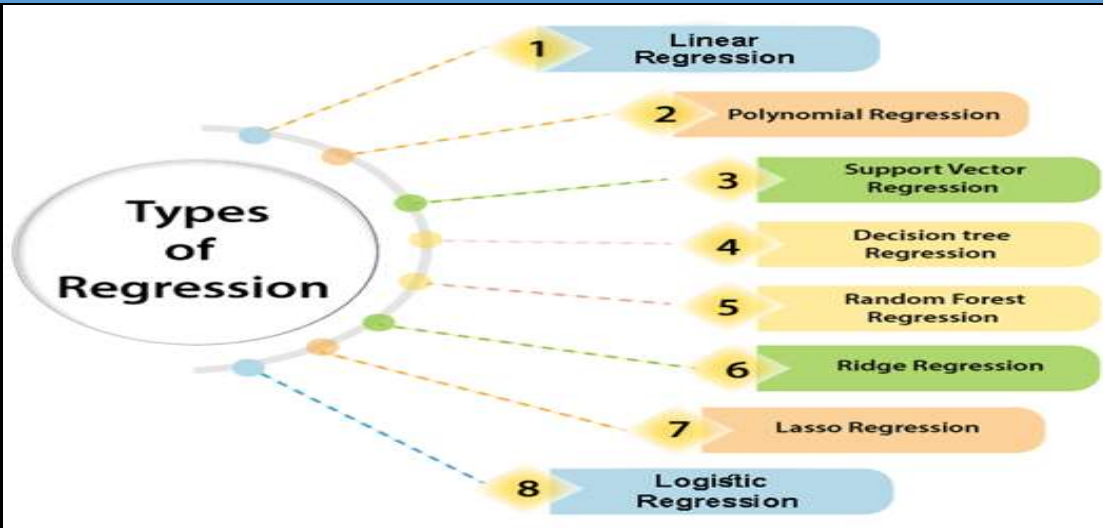
- Correlation – Scatter plots – Correlation coefficient for quantitative data – Computational formula for correlation coefficient
- Regression – A regression line – Least squares regression line – Standard error of estimate – Interpretation of r^2 – Multiple regression equations – Regression toward the mean.

REGRESSION

REGRESSION

- Regression → a statistical technique used to evaluate the **relationship** between two or more independent variables.
- Used by organizations to understand the significance of their data points and to make better **decisions** using analytical techniques.
- Thus if two variables are correlated, description can lead to **prediction**.
- Predictive **accuracy** increases with the strength of the underlying correlation.
- But if **misinterpreted** as a real effect, regression toward the mean can lead to erroneous conclusions.

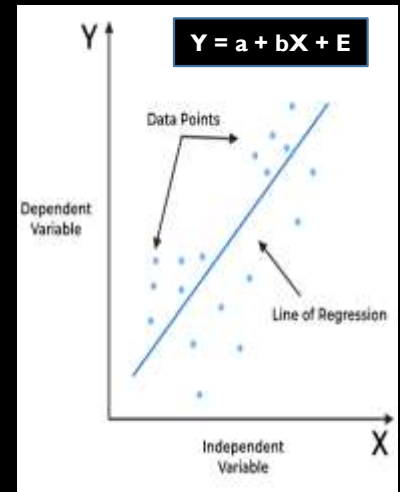
REGRESSION



A REGRESSION LINE

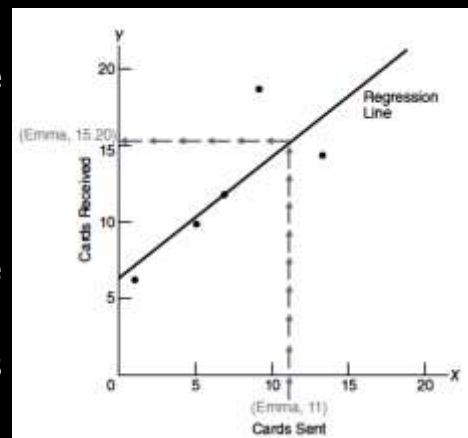
A REGRESSION LINE

- A regression line indicates a linear relationship between the **dependent** variables on the y-axis and the **independent** variables on the x-axis.
- X-axis → independent variable (eg. knowledge and IQ)
- Y-axis → dependent variable (eg. marks scored in exams)
- The correlation is established by analyzing the data pattern formed by the variables.
- The regression line is plotted closest to the data points in a regression graph.



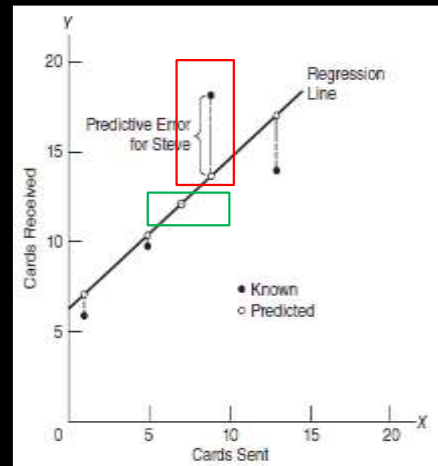
A REGRESSION LINE

- **Placement of line** – an important action.
- If all dots had defined a single straight line → placement of the regression line would have been simple → let it pass through all dots.
- Otherwise, placement of the regression line represents a compromise – adjustment.
- It passes through the main cluster, misses the other points.



A REGRESSION LINE

- **Prediction error** – an important factor
- The difference between actual value and the predicted value (obtained by the regression line)
- Solid dots → actual points; white dots → predicted points
- Red box shows the largest predictive error.
- Green box shows the smallest predictive error.
- The total predictive error (total of the vertical discrepancies between the actual and predicted dots) must be minimum.
- So it is desirable for the regression line to be placed in a position that **minimizes the total predictive error**



LEAST SQUARES REGRESSION LINE

LEAST SQUARES REGRESSION LINE

- Predictive errors may cause arithmetic standoff of zero (because of +ve and -ve error values)
- Hence total square predictive error is considered and it has to be minimized.
- The placement of the regression line minimizes the total square predictive error – that regression line is termed as least squares regression line.
- The original aim of using this LSR line → to minimize the total predictive error.
- Line created as a mathematical solution – even then only a close approximation to the LSR line is possible

LEAST SQUARES REGRESSION EQUATION

- The equation that minimizes the total of all squared prediction errors for known Y scores in the original correlation analysis.
- Y' → predicted value
- X → known value
- a, b → y-intercept and slope of the line → values calculated from the original correlation analysis

$$Y' = a + bX$$

$$SS_x = \sum (X - \bar{X})^2 = \sum X^2 - \frac{(\sum X)^2}{n}$$

$$SS_y = \sum (Y - \bar{Y})^2 = \sum Y^2 - \frac{(\sum Y)^2}{n}$$

$$b = r \sqrt{\frac{SS_y}{SS_x}}$$

$$a = \bar{Y} - b\bar{X}$$

r → the correlation between X and Y
 SS_y → the sum of squares for all Y scores
 SS_x → the sum of squares for all X scores
 $\bar{X}$ → the sample means for all X
 $\bar{Y}$ → the sample means for all Y

LEAST SQUARES REGRESSION EQUATION - EXAMPLE

Find the LSR eqn for the following data and use it to predict the value for $X=11$:

NAME	CARDS SENT X	CARDS RECEIVED Y
DD	13	14
SS	9	18
MM	7	12
AA	5	10
JJ	1	6

1. Find n.
2. Find r, SS_x, SS_y, b
3. Find $\bar{X}, \bar{Y}, a$

$$SS_x = 80 \quad SS_y = 80 \quad r = 0.8$$

$$b = 0.8 \quad a = 6.4$$

$$Y' = bX + a$$

$$Y' = 0.8X + 6.4$$

Find the predicted value for $X = 11$.

$$Y' = 15.20$$

LEAST SQUARES REGRESSION EQUATION

- Property of the least squares regression equation $\rightarrow$ it automatically minimizes the total of all squared predictive errors for known Y scores in the original correlation analysis.
- For new values of X, the value of Y' can be computed.
- Thus used to forecast an unknown event based on information about some known event. (Eg. – in previous slide)
- Because of the value of a, $Y' = a$, when $X=0$.
- Because of the value of b, Y' will show increments (when $b > 0$)
- Generate predictions from equations rather than graphs $\rightarrow$ more accurate and easier to generate predictions from equations.

$$Y' = bX + a$$

STANDARD ERROR OF ESTIMATE

STANDARD ERROR OF ESTIMATE

- Although a value is predicted using LSR equation, the actual Y may be different than Y' .
- LSR equation minimizes the predictive error, but does not eliminate the error.
- The amount of error associated with our predictions must be estimated.
- The smaller the estimated error $\rightarrow$ the better the prognosis (forecast) will be for the predictions
- Error estimation for new predictions indicates failure in the prediction of original data.
- Standard error of estimate $\rightarrow$ A rough measure of the average amount of predictive error

STANDARD ERROR OF ESTIMATE

- Key property → It is a special kind of standard deviation that reflects the magnitude of predictive error (average of $Y - Y'$)

STANDARD ERROR OF ESTIMATE

$$s_{y|x} = \sqrt{\frac{SS_{y|x}}{n-2}} = \sqrt{\frac{\sum(Y - Y')^2}{n-2}} = \sqrt{\frac{SS_y(1-r^2)}{n-2}}$$

- $SS_{y|x}$ → sum of the squares for predictive errors.
- $n-2$ → degrees of freedom –the last 2 values estimated must depend on remaining $n-2$ values for any straight (regression) line
- SS_y → sum of the squares for Y scores
- r → correlation coefficient

$$SS_y = \sum(Y - \bar{Y})^2 = \sum Y^2 - \frac{(\sum Y)^2}{n}$$

STANDARD ERROR OF ESTIMATE

- The standard error of estimate can be used as: $Y' = Y' \pm s_{y|x}$
- Eg.

NAME	CARDS SENT X	CARDS RECEIVED Y
DD	13	14
SS	9	18
MM	7	12
AA	5	10
JJ	1	6

$$Y' = bX + a$$

$$Y' = 0.8X + 6.4$$

$$Y' = 15.20$$

$$s_{y|x} = 3.1$$

$$Y' = 15.20 \pm 3.1$$

STANDARD ERROR OF ESTIMATE

STANDARD ERROR OF ESTIMATE

$$s_{y|x} = \sqrt{\frac{SS_{y|x}}{n-2}} = \sqrt{\frac{\sum(Y - Y')^2}{n-2}} = \sqrt{\frac{SS_y(1-r^2)}{n-2}}$$

Importance of r

- Perfect relationship $\rightarrow r = -1$ or $+1 \rightarrow$ Then $SS_{y|x} = 0 \rightarrow$ no predictive errors.
- No relationship $\rightarrow r = 0 \rightarrow$ Then $SS_{y|x} = SS_y \rightarrow$ highest predictive error possible.
- Thus prediction is favorable for perfect or strong relationships

INTERPRETATION OF R^2

INTERPRETATION OF R^2

- $r^2 \rightarrow$ squared correlation coefficient
 - Provides interpretation of the correlation coefficient
 - A measure of the predictive accuracy
- 2 kinds of predictive errors:
 - Errors due to the repetitive prediction of the mean
 - Errors due to the regression equation

INTERPRETATION OF R^2 REPETITIVE PREDICTION OF THE MEAN

- In $Y = aX + b$, if X is not known, Y' (predicted value) cannot be computed.
- But $\bar{Y}$ can be computed and used for predicting X
- For different values predicted for Y , repetitive prediction of different $\bar{Y}$ can be made
- This will provide a frame of reference against which the predictive effort based on correlation can be evaluated
- Smaller error variability $\rightarrow$ more accurate predictions of Y

INTERPRETATION OF R^2 PREDICTIVE ERRORS

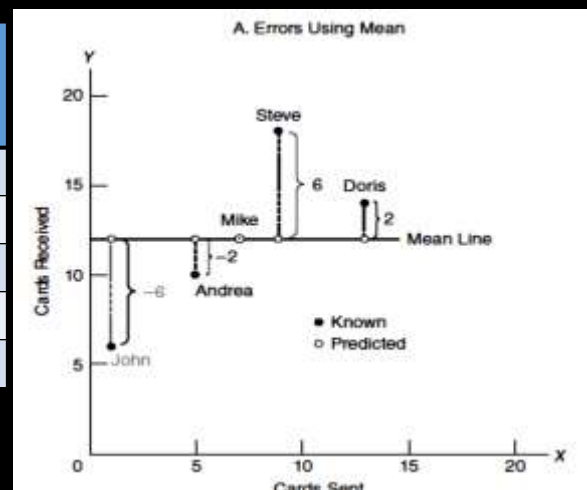
- 2 predictive efforts – for finding $\bar{Y}$ that can lead to predictive errors
- $\bar{Y}$ can be used when X is not available, and predictive errors can be calculated from mean
- Predictions of Y' from the least squares equation can be made and those Y' can have predictive errors

INTERPRETATION OF R^2 PREDICTIVE ERRORS

NAME	CARDS SENT X	CARDS RECEIVED Y
Andrea	5	10
Mike	7	12
Doris	13	14
Steve	9	18
John	1	6

$$\bar{Y} = 12$$

$\bar{Y}$ can be used for predicting errors for Y



INTERPRETATION OF R^2 PREDICTIVE ERRORS

$$b = r \sqrt{\frac{SS_y}{SS_x}}$$

$$a = \bar{Y} - b\bar{X}$$

$$Y' = bX + a$$

$$SS_x = \sum (X - \bar{X})^2 = \sum X^2 - \frac{(\sum X)^2}{n}$$

$$SS_y = \sum (Y - \bar{Y})^2 = \sum Y^2 - \frac{(\sum Y)^2}{n}$$

$$SS_x = 80 \quad SS_y = 80 \quad r = 0.8$$

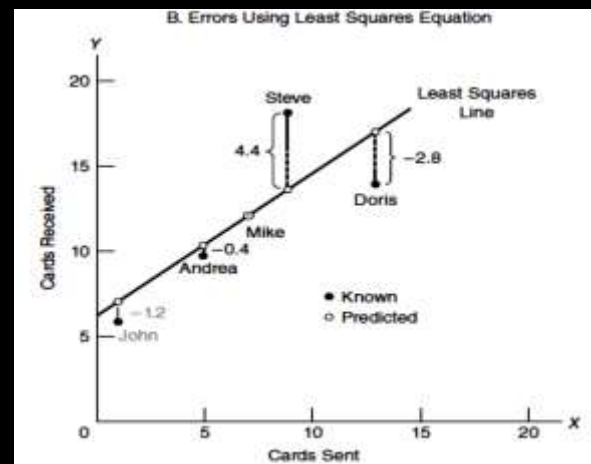
$$b = 0.8 \quad a = 6.4$$

$$Y' = 0.8X + 6.4$$

NAME	CARDS SENT X	CARDS RECEIVED Y	Predicted Cards Received Y'
Andrea	5	10	10.4
Mike	7	12	12
Doris	13	14	17.8
Steve	9	18	13.6
John	1	6	7.2

INTERPRETATION OF R^2 PREDICTIVE ERRORS

NAME	CARDS SENT X	CARDS RECEIVED Y	Predicted Y'
Andrea	5	10	10.4
Mike	7	12	12
Doris	13	14	17.8
Steve	9	18	13.6
John	1	6	7.2



Y' found using Least Squares equation can be used for predicting errors for Y

INTERPRETATION OF R^2 PREDICTIVE ERRORS

- For Steve, when $\bar{Y}=12$ is used to predict the score for Y, **error** = $Y - \bar{Y} = 18 - 12 = 6$
- But when Y' is used to predict the score for Y, **error** = $Y - Y' = 18 - 13.6 = 4.4$
- Overall, errors are smaller when customized predictions of Y' from the least squares equation can be used (because X scores are known) than when only the repetitive prediction of Y can be used (because X scores are not known).

NAME	CARDS SENT X	CARDS RECEIVED Y	Predicted Y'
Andrea	5	10	10.4
Mike	7	12	12
Doris	13	14	17.8
Steve	9	18	13.6
John	1	6	7.2

$$\bar{Y} = 12$$

INTERPRETATION OF R^2 ERROR VARIABILITY (SUM OF SQUARES)

- Which predictive effort is accurate – using mean or using Y' ?
- Evaluated using a measure of the collective errors produced by each effort → using sum of squares → The deviations (or) errors are squared and then summed up.
- The error variability for the repetitive prediction of the

mean $SS_y = \sum(Y - \bar{Y})^2$

- The error variability for the customized predictions from the least squares equation

$$SS_{y|x} = \sum(Y - Y')^2$$

- For this dataset, $SS_y = 80$, $SS_{y|x} = 28.8$

NAME	CARDS SENT X	CARDS RECEIVED Y	Predicted Cards Received Y'
Andrea	5	10	10.4
Mike	7	12	12
Doris	13	14	17.8
Steve	9	18	13.6
John	1	6	7.2

INTERPRETATION OF R^2 PROPORTION OF PREDICTED VARIABILITY

- $SS_y \rightarrow$ measures the **total variability** of Y scores $\rightarrow$ based on $\bar{Y}$ (and no X scores)
- $SS_{y|x} \rightarrow$ measures the **residual variability** Y scores $\rightarrow$ based on least squares (and using X scores)
- SS measure of the actual gain in accuracy due to the least squares predictions = total variability – residual variability = variability predictable from the regression equation, **$SS_{y'} = SS_y - SS_{y|x} = 80 - 28.8 = 51.2$**
- Gain in accuracy relative to the original error variability

$$= \frac{SS_y - SS_{y|x}}{SS_y} = \frac{80 - 28.8}{80} = 0.64$$
- Percentage gain in prediction accuracy = 64% $\rightarrow$ the proportion or percent of the total variability of SS_y that is predictable from its relationship with the X variable.

INTERPRETATION OF R^2 PROPORTION OF PREDICTED VARIABILITY

- Gain in prediction accuracy = 0.64
- For the used dataset, $r = 0.8$, and hence $r^2 = 0.64 \rightarrow$ means 64% can be predicted correctly
- Hence $r^2 \rightarrow$ also called as proportion of predictable variability $\rightarrow$ provides with a measure of the worth of least squares predictive effort.

The square of the correlation coefficient, r^2 , *always* indicates the proportion of total variability in one variable that is predictable from its relationship with the other variable.

- $$r^2 = \frac{SS_{y'}}{SS_y} = \frac{SS_y - SS_{y|x}}{SS_y} = \frac{\sum (y' - \bar{y})^2}{SS_y}$$

INTERPRETATION OF R^2 PROPORTION OF PREDICTED VARIABILITY

- All formulae

$$SS_x = \sum (X - \bar{X})^2 = \sum X^2 - \frac{(\sum X)^2}{n}$$

$$SS_y = \sum (Y - \bar{Y})^2 = \sum Y^2 - \frac{(\sum Y)^2}{n}$$

$$b = r \sqrt{\frac{SS_y}{SS_x}}$$

$$a = \bar{Y} - b\bar{X}$$

$$Y' = bX + a$$

$$SS_y = \sum (Y - \bar{Y})^2$$

$$SS_{y|x} = \sum (Y - Y')^2$$

$$SS_{y'} = SS_y - SS_{y|x} = \sum (Y' - \bar{Y})^2$$

$$r^2 = \frac{SS_{y'}}{SS_y} = \frac{SS_y - SS_{y|x}}{SS_y}$$

INTERPRETATION OF R^2

r^2 Does Not Apply to Individual Scores

- Applied to the variability for the entire set of Y scores
- $r^2 = 64\%$ represents a reduction in the **total variability** for the five Y scores when they are replaced by a Y', given the least squares equation ($Y' = aX + b$) and various values of X.

Small Values of r^2

- Unlike r, even smaller values of r^2 will help to measure the variability for a single variable.

r^2 Doesn't Ensure Cause-Effect

- r^2 indicates only the proportion or percent of predictable variability, and not causality

MULTIPLE REGRESSION EQUATIONS

MULTIPLE REGRESSION EQUATIONS

- Complex equation that contains not just one but several X (predictor variables) and one Y' (criterion variable). Eg. $Y' = a_1x_1 + a_2x_2 + a_3x_3 + \dots a_nx_n + b$

$$Y' = 0.410(X_1) + 0.005(X_2) + 0.001(X_3) + 1.03$$

- Supply more accurate predictions for Y'

Features

- Similar to simple regression equation, multiple regression equation
 - Minimizes the sum of the squared predictive errors.
 - Has standard errors of estimate that roughly measure the average amounts of predictive error

REGRESSION TOWARD THE MEAN

REGRESSION TOWARD THE MEAN

- Regression toward the mean → Tendency for scores, particularly extreme scores, to shrink toward the mean.
- Eg. In test-1, for the top scorers, which happened because of true effort (or) because of good luck / chances. In test-2, their scores may reduce or move towards the mean

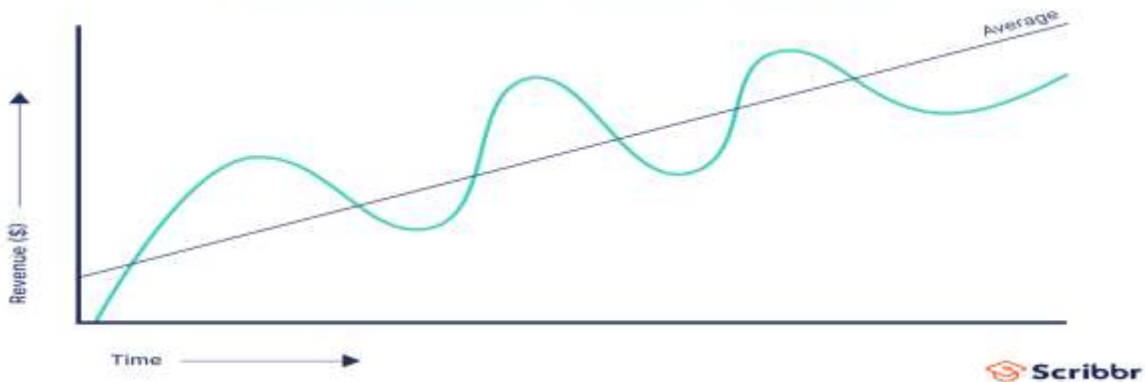
Appears in Many Distributions

- Regression toward the mean appears among subsets of extreme observations for a wide variety of distributions.
- Observed regression toward the mean occurs for individuals or subsets of individuals, not for entire groups

REGRESSION TOWARD THE MEAN

Regression to the mean

A statistical phenomenon that describes how variables with outliers on the first measurement are often closer to the mean when measured again.



REGRESSION TOWARD THE MEAN

The Regression Fallacy

- Fallacy → wrong notion
- Committed whenever regression toward the mean is interpreted as a real, rather than a chance, effect.
- 1st time winners praised, losers reprimanded (scolded)
- 2nd time, the praised winners didn't perform well, and the losers performed better
- A conclusion → praise hinders but a reprimand helps performance
- Everything happens and chance plays a role – which cannot be predicted

Avoiding The Regression Fallacy

- The regression fallacy can be avoided by splitting the subset of extreme observations into two groups.
- Research with underachievers always includes a control group for regression toward the mean.